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Set-2

END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering

Semester: V

Name of the Paper: Petroleum Geophysics

Course Code: TPE-501

Time: Three Hours

MM: 100

Note:

- All questions are compulsory.
- Answer any two sub questions in each main question.
- Total marks for each main question is **twenty**
- Each Question carries **10 marks**

Q1		COs
a.	Explain the role Geophysics in petroleum and mining Industry? What are the various earth properties are measured in Geophysics?	CO1
b.	How is Geophysics used for imaging the subsurface structure? Explain contour map with diagram.	CO1 CO2
c.	What are the passive and active method of geophysics? What is the objective of geophysical investigations?	CO1 CO2
Q2		
a.	How many geophysical methods are available for oil exploration? Explain the Gravity and magnetic survey method in details?	CO2
b.	What are Electromagnetic methods of geophysical survey? Where this survey is useful?	CO2
c.	Explain Electrical methods for oil exploration. What is the data format structure of Seismic data?	CO3
Q3		
a.	What is seismic method for oil & gas exploration? What are the parameters measured in this method?	CO3
b.	Why Seismic is widely accepted in oil exploration? What is the principle of Geophones?	CO3
c.	Explain Common Depth Point (CDP) Reflection Technique by drawing the figure. Write formula for foldage.	CO3
Q4		
a.	What is seismic reflection prospecting methodology explain by drawing a figure? Write the formula for reflection coefficient.	CO4
b.	What is the various correction applied in reflection and refraction during the seismic data acquisition?	CO4
c.	What is well to seismic tie? Explain with figure? Why it is required?	CO4

Q5		
a.	How is the seismic trace generated? Explain Convolution model of seismic trace.	CO5
b.	How VSP data is recorded? Why is it required? What is the application of VSP data?	CO5
c.	What is T-D relationship curve and how it is generated? Why it is important? How well to seismic tie is performed?	CO5